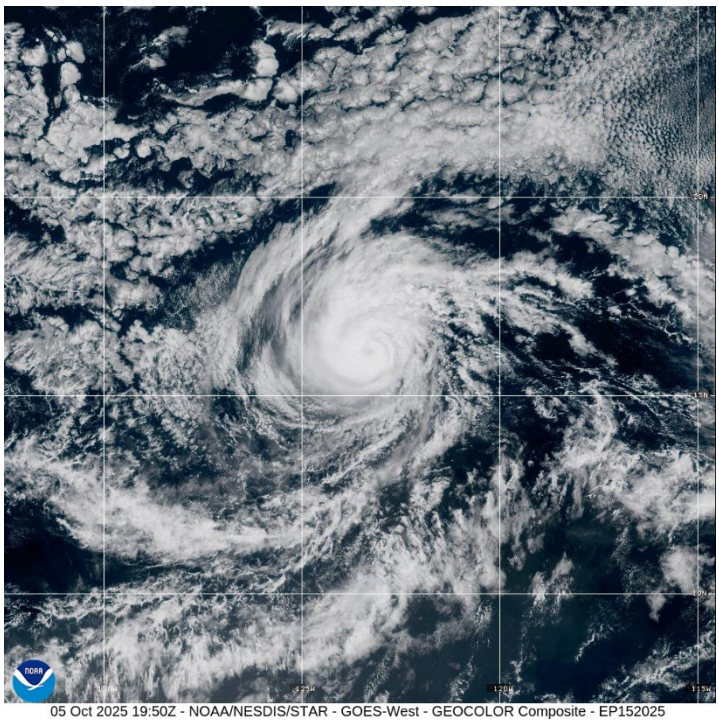
NATIONAL HURRICANE CENTER TROPICAL CYCLONE REPORT[1]

HURRICANE OCTAVE

(EP152025)

30 September–9 October 2025

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National Hurricane Center
3 December 2025



GOES-18 GEOCOLOR IMAGE OF HURRICANE OCTAVE AT 1950 UTC 05 OCTOBER 2025. IMAGE COURTESY OF NOAA/NESDIS/STAR.

Octave was a category 1 hurricane (on the Saffir-Simpson Hurricane Wind Scale) that remained away from land over the central portion of the eastern Pacific.

[1] This is an abbreviated Tropical Cyclone Report since there were no coastal watches or warnings issued and no direct fatalities reported in association with Octave.

Hurricane Octave

30 SEPTEMBER–9 OCTOBER 2025

BEST TRACK

Hurricane Octave took a rather unusual track across the eastern Pacific as the cyclone was highly influenced by two other tropical cyclones, Hurricane Priscilla and Tropical Storm Raymond (Fig. 1). After tracking steadily west-northwestward for several days, Octave made an abrupt U-turn on 5 October. The cyclone was steered eastward beginning that day by an approaching upper-level trough to the northwest along with the breakdown of a subtropical ridge to the east of the cyclone, caused by the large circulation of Hurricane Priscilla. Octave was then influenced by the circulations of both Priscilla and Raymond during the next several days, as it tracked generally eastward and eventually dissipated by 10 October. The "best track[2]" positions and intensities for Hurricane Octave are listed in Table 1. The best track chart of Octave's path is given in Fig. 2, with the wind and pressure histories along with available observations[3] shown in Figs. 3 and 4, respectively.

There were no surface observations of winds of tropical storm force associated with Octave.

Origin

The origin of Octave can be traced to a convective complex that developed over the higher terrain of the far northern portion of South America on 16–17 September, fueled by an influx of deeper moisture feeding in off the tropical Atlantic from the east. The disturbance moved westward across Venezuela and Colombia between 17 and 19 September before emerging off the Colombian coast on 20 September. The disturbance then crossed Panama and Costa Rica on 21–22 September and moved offshore of Central America by 23 September. The disturbance continued steadily west-southwestward over the following several days, and a small area of low pressure developed several hundred miles off the southwestern coast of Mexico on 29 September. Convection gradually increased and became better organized, and the system

[2] A digital record of the complete best track, including wind radii, can be found on line at ftp://ftp.nhc.noaa.gov/atcf. Data for the current year's storms are located in the *btk* directory, while previous years' data are located in the *archive* directory.

[3] Observations include subjective satellite-based Dvorak technique intensity estimates from the Tropical Analysis and Forecast Branch (TAFB) and the Satellite Analysis Branch (SAB), objective Advanced Dvorak Technique (ADT) estimates and Satellite Consensus (SATCON) estimates from the Cooperative Institute for Meteorological Satellite Studies/University of Wisconsin-Madison. Data and imagery from NOAA polar-orbiting satellites including the Advanced Microwave Sounding Unit (AMSU), the NASA Global Precipitation Mission (GPM), the European Space Agency's Advanced Scatterometer (ASCAT), the Defense Meteorological Satellite Program (DMSP) satellites, and the Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (TROPICS) satellites, among others, were also useful in constructing the best track of Octave.

became a tropical depression by 0000 UTC 30 September, about 825 n mi south of the southern tip of the Baja California Peninsula.

Peak Intensity and Minimum Pressure

Octave's peak intensity of 80 kt is based on a subjective Dvorak current intensity estimate of T4.5 from SAB, as well as objective intensity estimates from UW-CIMSS that ranged from 75 to 79 kt at 0000 UTC 6 October. An internal subjective Dvorak analysis using the eye pattern technique yielded a data T-number of 5.0 for roughly half of the 10-minute GOES-18 satellite images between 2100 UTC 5 October and 0000 UTC 6 October. All of these data were considered when assigning the peak intensity for Octave. A 0209 UTC 6 October Radarsat Constellation Mission-2 (RCM-2) Synthetic Aperture Radar (SAR) pass indicated a well-defined structure to the cyclone, with peak winds in the northwest quadrant of Octave's circulation (Fig.5).

The estimated minimum central pressure of 980 mb is based on the Knaff-Zehr-Courtney pressure-wind relationship for a storm of Octave's intensity and size.

CASUALTY AND DAMAGE STATISTICS

There were no reports of damage or casualties associated with Octave.

FORECAST AND WARNING VERIFICATION

The genesis forecasts for Octave's formation were poor, likely a result of the global models struggling to resolve what would eventually be three tropical cyclones in close proximity to each other off the western coast of Mexico (Fig. 1). The disturbance from which Octave developed was first highlighted in the Tropical Weather Outlook (TWO) 60 h prior to formation (Table 2). The 7-day genesis probability was raised to the medium category 42 h before formation and to the high category 30 h in advance. Two-day genesis forecasts were a little better; low probabilities were introduced 60 h prior to formation, medium probabilities 30 h prior, and high probabilities 18 h before development occurred.

Octave's genesis location was on the south side of all genesis formation regions. However, the system formed within each of the genesis forecast areas depicted in NHC's Graphical TWO, resulting in a 100% hit rate (Fig. 6).

A verification of NHC official track forecasts for Octave is given in Table 3a. Official track forecast errors were lower than the 5-year means at all forecast intervals. A homogeneous comparison of the official track errors with selected guidance models is given in Table 3b. The official track forecasts outperformed nearly all guidance from 12 h through 72 h. Several models outperformed the official track forecasts at 96 h and 120 h. Notably, the Google DeepMind Ensemble Mean (GDMI) outperformed the official track forecasts at all forecast hours, except the 12-h interval.

A verification of NHC official intensity forecasts for Octave is given in Table 4a. Official intensity forecast errors were lower than the mean official errors for the previous 5-year period at all forecast intervals. A homogeneous comparison of the official intensity errors with selected guidance models is given in Table 4b. The official intensity forecasts outperformed nearly all of the intensity guidance. The exception again was the GDMI, which outperformed the official intensity forecasts at many forecast intervals.

There were no coastal watches or warnings issued for Octave.

ACKNOWLEDGEMENTS

Dr. Philippe Papin (NHC) created the GTWO verification graphic (Figure 6).

Table 1. Best track for Hurricane Octave, 30 September – 9 October 2025.

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
29 / 1800	9.3	112.2	1008	30	low
30 / 0000	9.3	112.4	1008	30	tropical depression
30 / 0600	9.3	112.9	1007	30	"
30 / 1200	9.5	113.3	1005	35	tropical storm
30 / 1800	9.9	113.8	1005	35	"
01 / 0000	10.3	114.3	1003	40	"
01 / 0600	10.9	114.8	1002	45	"
01 / 1200	11.4	115.4	1000	50	"
01 / 1800	11.7	115.9	1000	50	"
02 / 0000	11.9	116.6	999	55	"
02 / 0600	12.0	117.4	999	55	"
02 / 1200	12.3	118.0	999	55	"
02 / 1800	12.6	118.7	999	55	"
03 / 0000	12.9	119.2	1001	50	"
03 / 0600	13.1	119.9	1003	45	"
03 / 1200	13.2	120.8	1003	45	"
03 / 1800	13.3	121.6	1001	50	"
04 / 0000	13.6	122.3	1001	50	"
04 / 0600	13.9	122.9	1001	50	"
04 / 1200	14.3	123.5	999	55	"
04 / 1800	14.7	123.9	999	55	"
05 / 0000	15.0	124.1	996	60	"
05 / 0600	15.4	124.2	992	65	hurricane
05 / 1200	15.6	123.9	987	70	"
05 / 1800	15.9	123.5	984	75	"
06 / 0000	16.2	123.1	980	80	"

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
06 / 0600	16.3	122.5	984	75	"
06 / 1200	16.3	121.9	993	60	tropical storm
06 / 1800	16.2	121.3	999	50	"
07 / 0000	15.9	120.7	1001	45	"
07 / 0600	15.8	120.2	1001	45	"
07 / 1200	15.7	119.7	1001	45	"
07 / 1800	15.5	119.2	1001	45	"
08 / 0000	15.3	118.5	1003	40	"
08 / 0600	15.2	117.8	1003	40	"
08 / 1200	15.2	116.9	1003	40	"
08 / 1800	15.5	115.8	1003	40	"
09 / 0000	15.9	114.0	1003	40	"
09 / 0600	16.6	112.4	1005	35	"
09 / 1200	17.6	111.0	1007	30	low
09 / 1800	18.0	109.6	1007	30	"
10 / 0000	-	-	-	-	dissipated
06 / 0000	16.2	123.1	980	80	maximum wind and minimum pressure

Table 2. Number of hours in advance of formation associated with the first NHC Tropical Weather Outlook forecast in the indicated likelihood category. Note that the timings for the “Low” category do not include forecasts of a 0% chance of genesis.

	Hours Before Genesis	
	48-Hour Outlook	**168-Hour Outlook**
Low (<40%)	60	60
Medium (40%-60%)	30	42
High (>60%)	18	30

Table 3a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) track forecast errors (n mi) for Hurricane Octave, 30 September – 9 October 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**17.4**	**27.6**	**38.6**	**47.3**	**53.2**	**62.2**	**97.6**	**108.4**
OCD5	35.9	81.5	132.6	195.0	271.8	358.5	509.4	624.1
Forecasts	35	33	31	29	27	25	21	17
OFCL (2020-24)	22.4	33.6	43.5	53.2	65.1	79.4	108.6	135.2
OCD5 (2020-24)	38.2	75.3	117.2	161.0	206.0	252.0	340.7	422.7

Table 3b. Homogeneous comparison of selected track forecast guidance models (in n mi) for Hurricane Octave, 30 September – 9 October 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 3a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	16.4	23.1	30.8	37.8	44.7	53.6	86.3	110.8
OCD5	38.2	93.0	158.0	243.6	339.8	443.4	619.3	735.0
GFSI	19.4	29.1	35.7	45.3	56.5	62.0	**65.0**	**81.5**
HWFI	23.0	37.5	59.0	90.3	137.0	183.7	228.5	244.3
HMNI	21.4	40.0	60.9	88.5	123.9	163.2	252.9	273.5
HFAI	20.7	34.9	51.9	75.6	99.4	123.4	151.1	179.1
HFBI	21.7	37.4	56.0	73.7	92.9	112.8	138.7	150.4
EMXI	17.5	27.9	34.4	39.8	54.3	64.4	**85.1**	**96.4**
CMCI	21.1	32.5	41.3	50.6	57.5	59.9	**71.0**	114.7
CTCI	20.2	35.0	57.3	85.8	112.3	129.2	169.7	243.2
TVCE	17.0	27.3	40.8	59.3	80.7	101.6	136.2	159.4
TVCX	17.2	26.6	39.6	56.3	76.7	97.0	128.3	147.3
TVDG	17.3	25.8	36.1	50.9	69.1	86.9	118.6	140.8
GFEX	16.6	25.8	31.7	38.5	48.7	56.3	**65.0**	**59.4**
HCCA	17.2	27.2	35.0	44.3	50.0	55.9	**67.7**	**83.1**
FSSE	16.9	27.0	33.1	41.6	46.2	55.5	**73.2**	**89.4**
AEMI	22.0	32.0	40.2	50.6	65.9	82.3	121.7	141.1
GDMI	17.6	**21.1**	**23.4**	**31.7**	**36.6**	**40.1**	**53.2**	**84.9**
TABS	29.6	58.9	91.9	113.2	118.9	122.0	142.2	188.2
TABM	24.4	42.4	53.7	64.7	81.9	103.9	131.1	122.7
TABD	32.2	67.5	96.9	119.9	141.5	167.4	192.2	147.1
Forecasts	27	26	24	22	20	18	14	10

Table 4a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) intensity forecast errors (kt) Hurricane Octave, 30 September – 9 October 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**4.0**	**7.7**	**8.7**	**9.3**	**8.1**	**6.0**	**7.6**	**9.1**
OCD5	5.7	9.6	12.2	15.0	15.2	15.6	10.8	11.5
Forecasts	35	33	31	29	27	25	21	17
OFCL (2020-24)	5.7	8.9	10.8	12.9	14.4	15.5	17.0	18.6
OCD5 (2020-24)	7.4	12.4	16.2	19.2	21.0	21.3	21.5	21.6

Table 4b. Homogeneous comparison of selected intensity forecast guidance models (in kt) for Hurricane Octave, 30 September – 9 October 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 4a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	4.1	7.9	9.0	9.8	8.2	5.6	7.5	7.5
OCD5	6.2	10.9	14.0	16.7	15.6	15.0	10.4	11.2
HWFI	8.7	11.5	13.1	14.6	16.2	13.5	11.4	8.7
HMNI	6.4	9.6	11.8	12.4	12.3	10.8	11.2	18.2
HFAI	7.1	9.5	13.1	13.7	16.2	17.3	11.1	**7.0**
HFBI	7.3	9.5	11.8	12.6	11.6	10.8	9.5	8.2
DSHP	5.8	9.0	11.1	11.6	10.7	9.2	13.1	11.4
LGEM	6.1	10.0	12.7	13.6	12.8	10.7	14.4	12.2
ICON	5.4	8.3	10.3	11.6	12.3	9.8	10.6	11.6
IVCN	5.1	8.0	10.2	10.6	11.1	8.8	8.9	10.3
IVDR	5.2	8.1	10.5	10.8	11.2	9.6	8.5	9.6
CTCI	**3.8**	**6.5**	9.5	11.2	11.2	12.1	18.6	20.0
GFSI	5.7	8.2	9.9	10.9	11.1	10.3	8.4	11.6
EMXI	6.2	10.2	12.8	14.3	14.6	14.4	19.9	20.6
HCCA	4.7	8.2	10.7	11.8	13.8	12.4	7.7	8.5
FSSE	5.0	7.9	10.4	10.2	10.1	8.1	12.5	15.6
CMCI	6.0	9.8	13.3	15.0	15.2	14.1	13.0	12.4
AEMI	6.7	9.9	12.0	13.5	14.4	13.7	14.5	12.4
GDMI	4.4	**5.7**	**5.2**	**5.8**	**5.9**	**4.9**	7.8	8.8
Forecasts	27	26	24	22	20	18	14	10

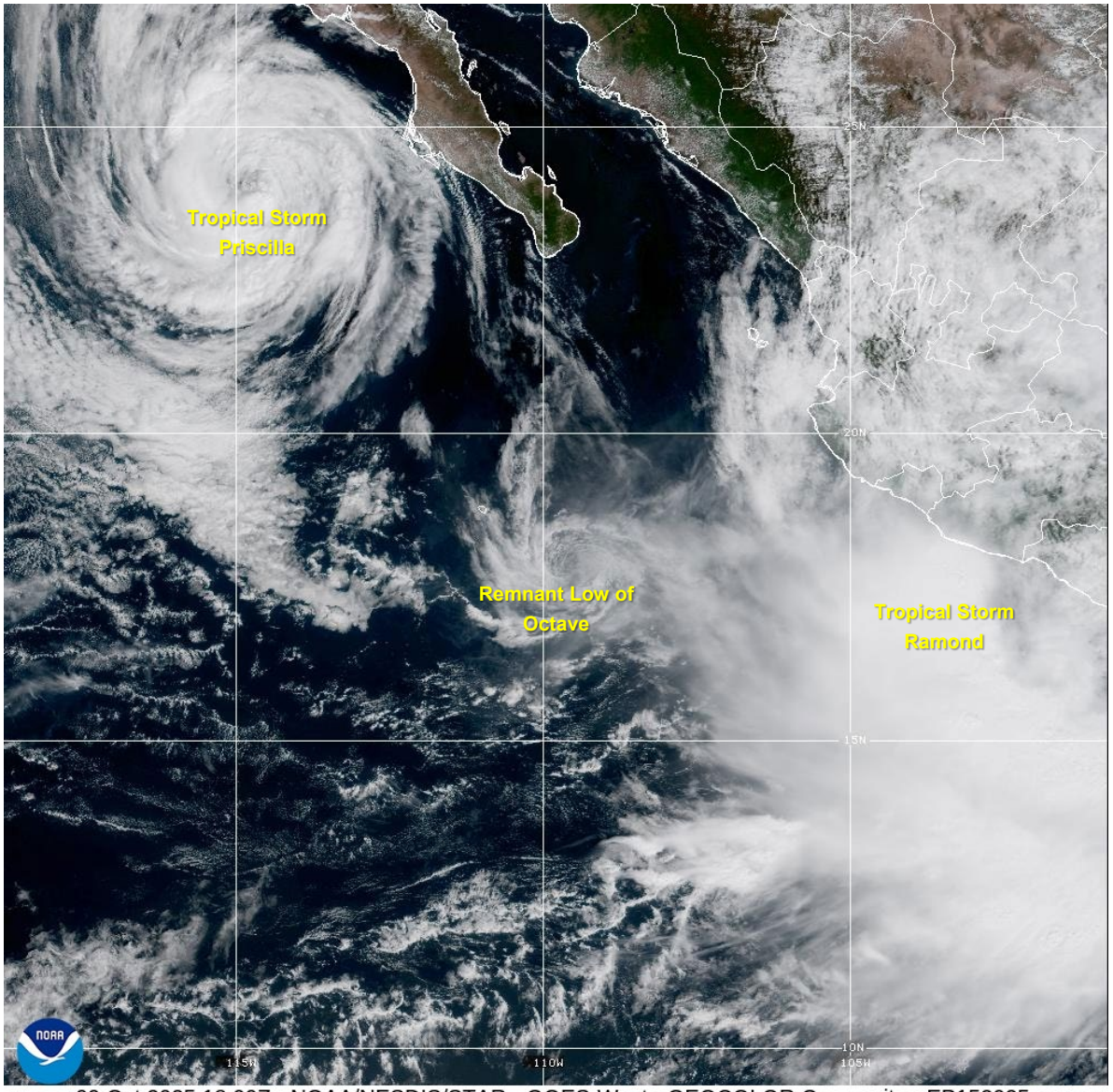


Figure 1. GOES-18 Geocolor image at 1800 UTC 9 October, showing Tropical Storm Priscilla, Tropical Storm Raymond, and the remnant low of former Hurricane Octave. Image courtesy of NOAA/NESDIS/STAR.

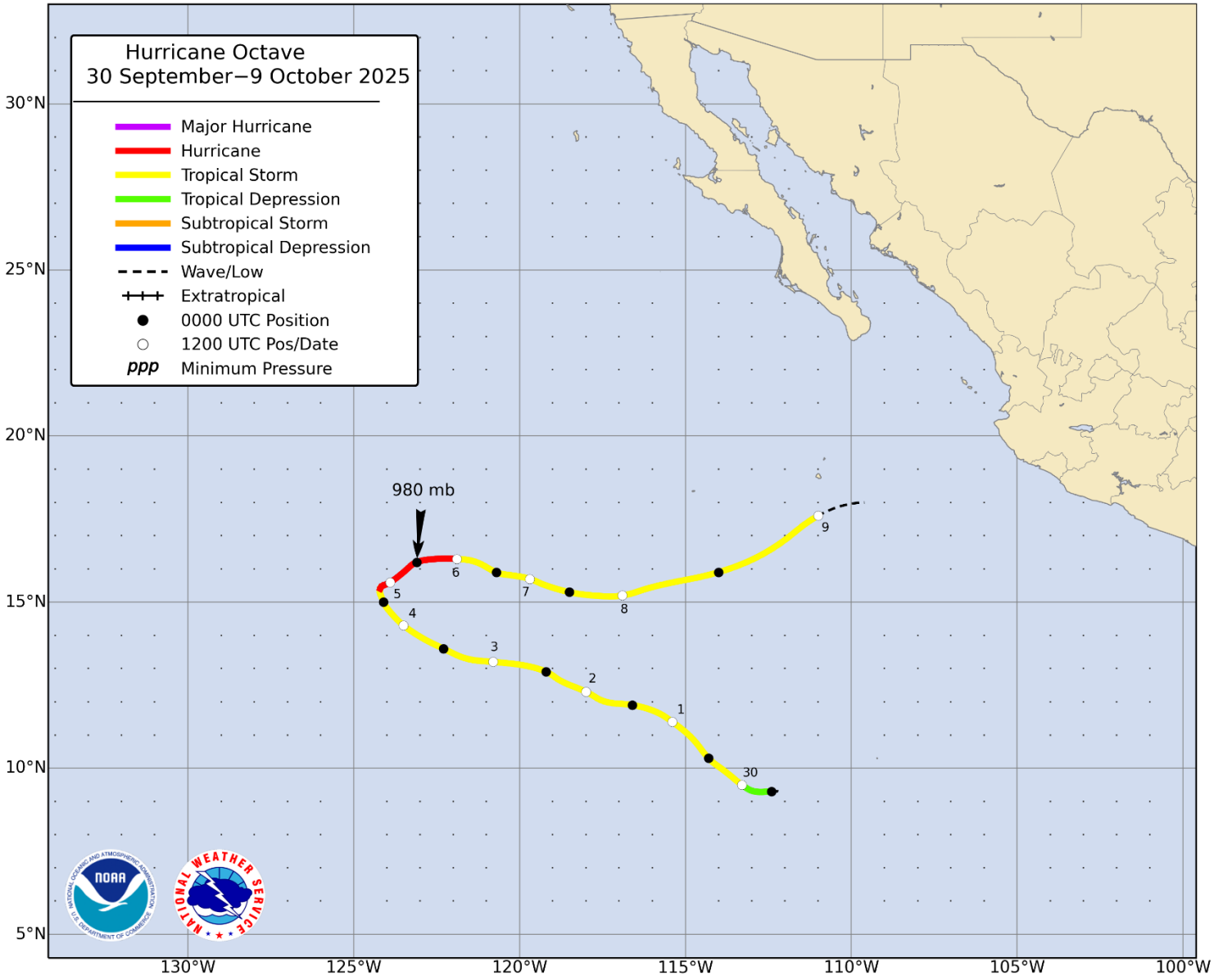


Figure 2. Best track positions for Hurricane Octave, 30 September – 9 October 2025.

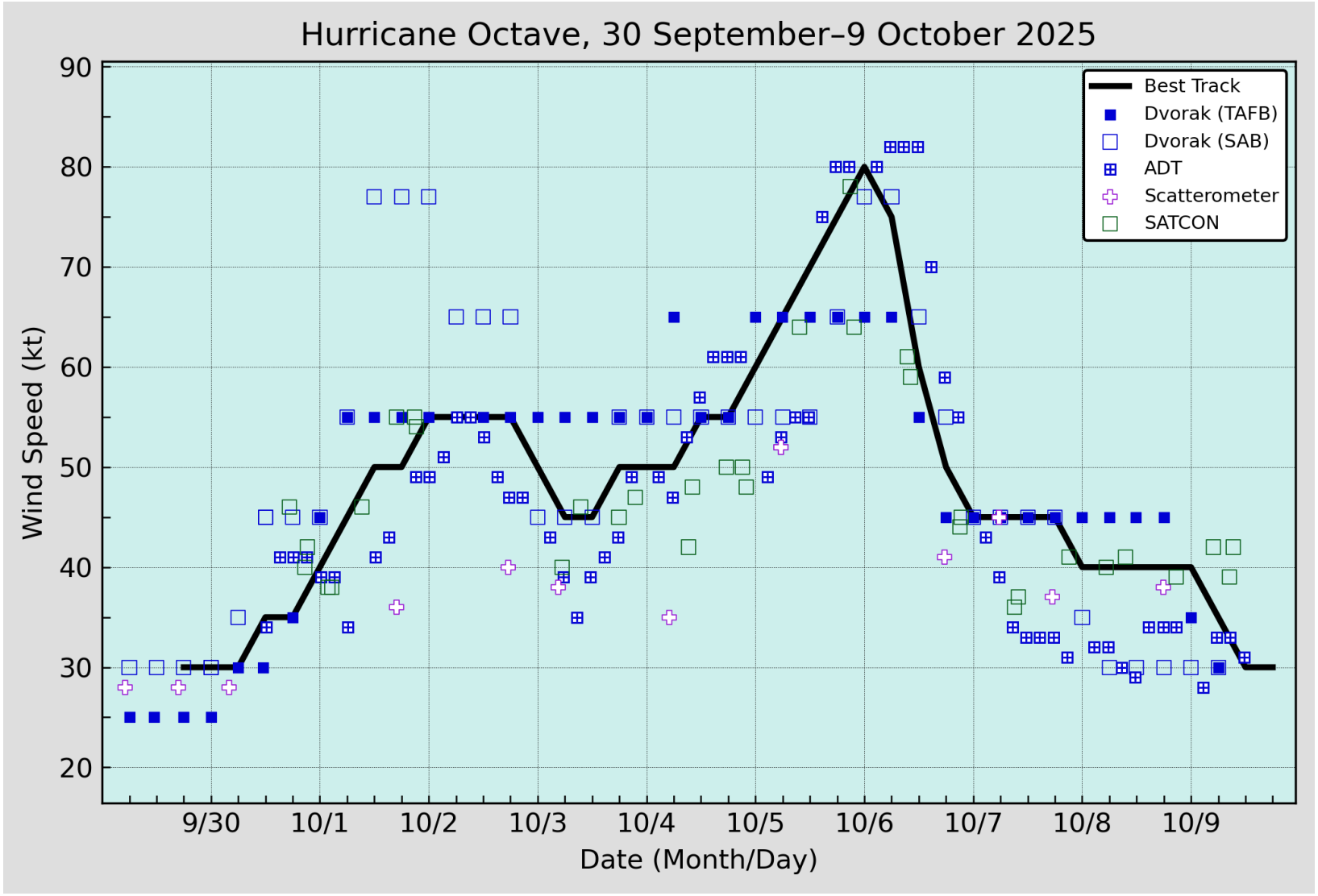


Figure 3. Selected wind observations and best track maximum sustained surface wind speed curve for Hurricane Octave, 30 September – 9 October 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. Dashed vertical lines correspond to 0000 UTC.

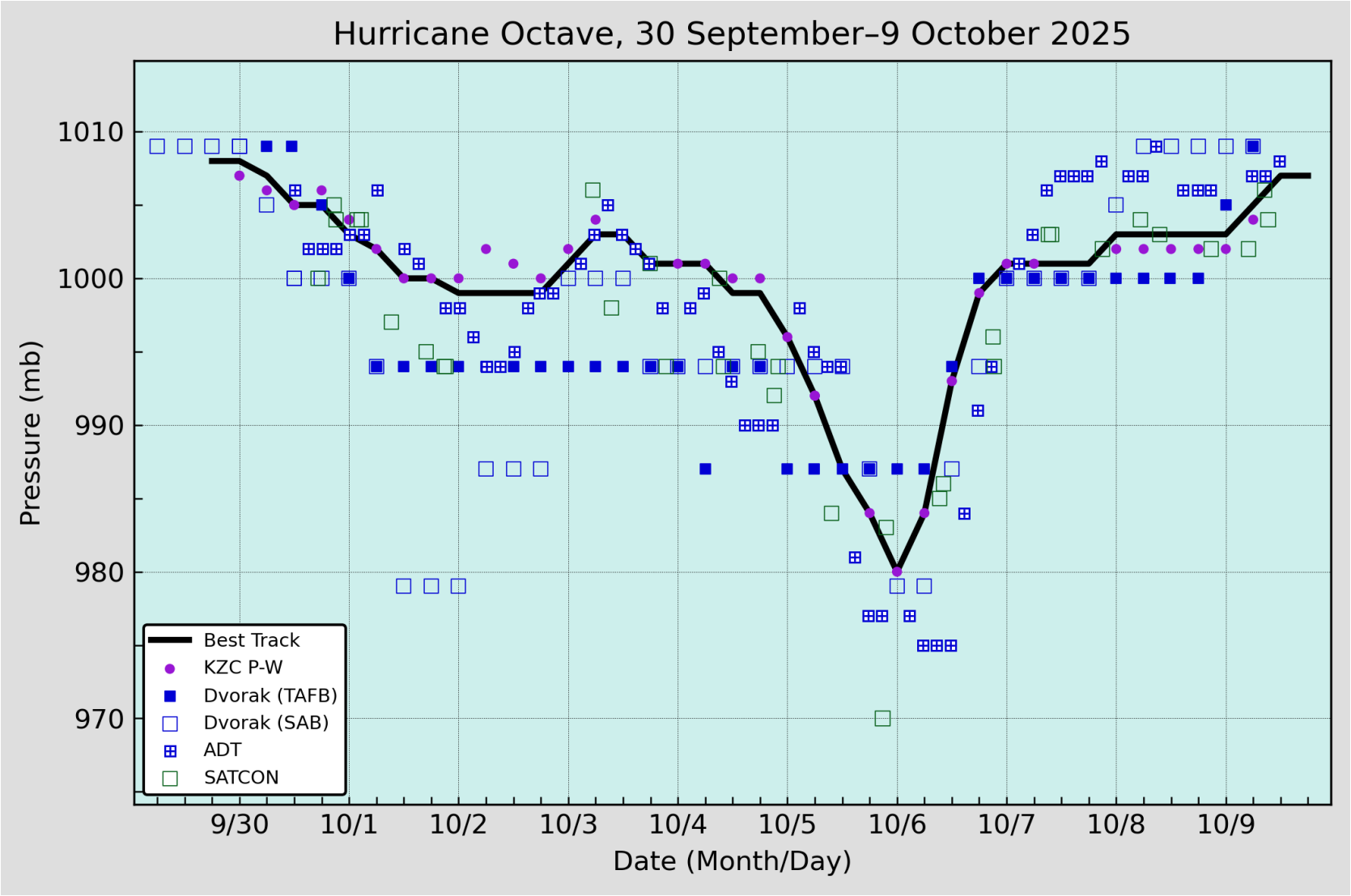


Figure 4. Selected pressure observations and best track minimum central pressure curve for Hurricane Octave, 30 September – 9 October 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. KZC P-W refers to pressure estimates derived using the Knaff-Zehr-Courtney pressure-wind relationship. Dashed vertical lines correspond to 0000 UTC.

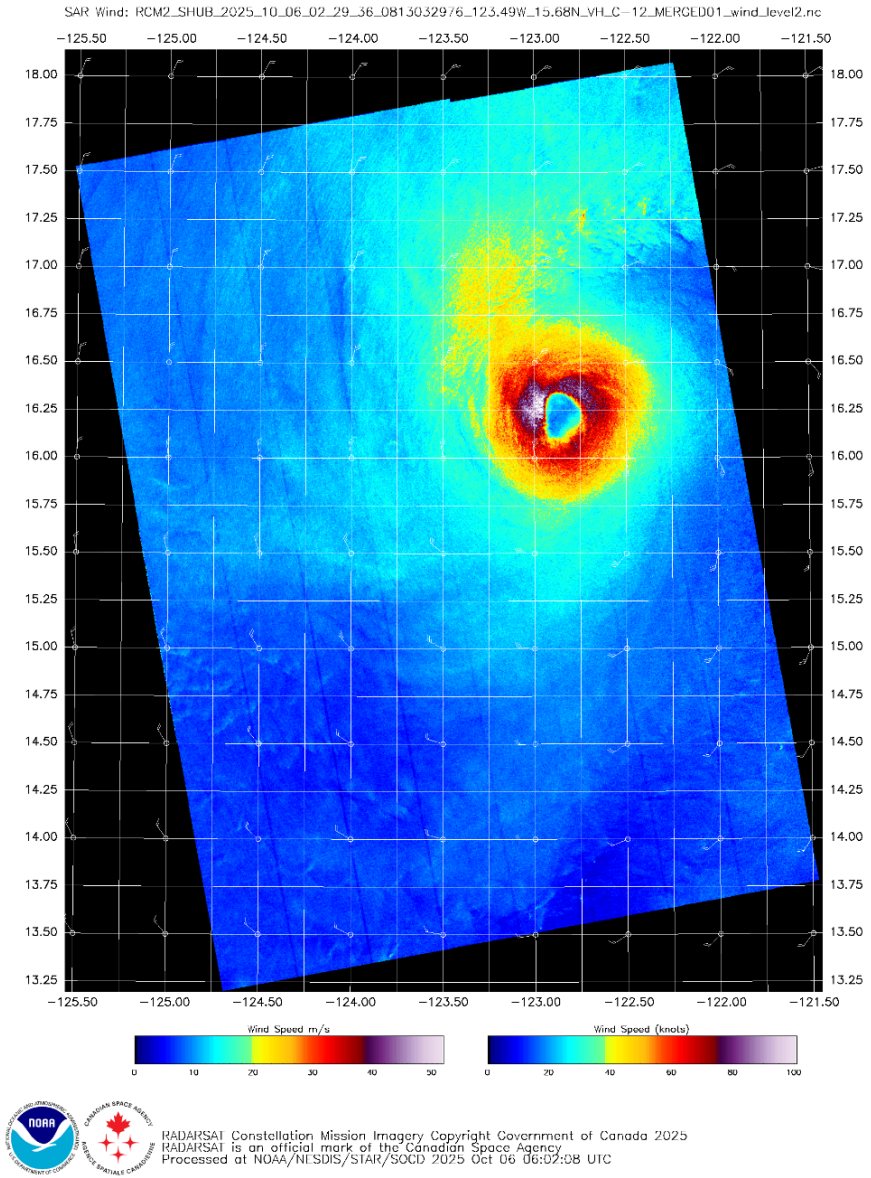


Figure 5. RCM-2 SAR microwave satellite image of Hurricane Octave 0229 UTC 6 October. Note: The validation of wind estimates from this instrument is still being examined, and the data were only used qualitatively when estimating Octave's peak intensity.

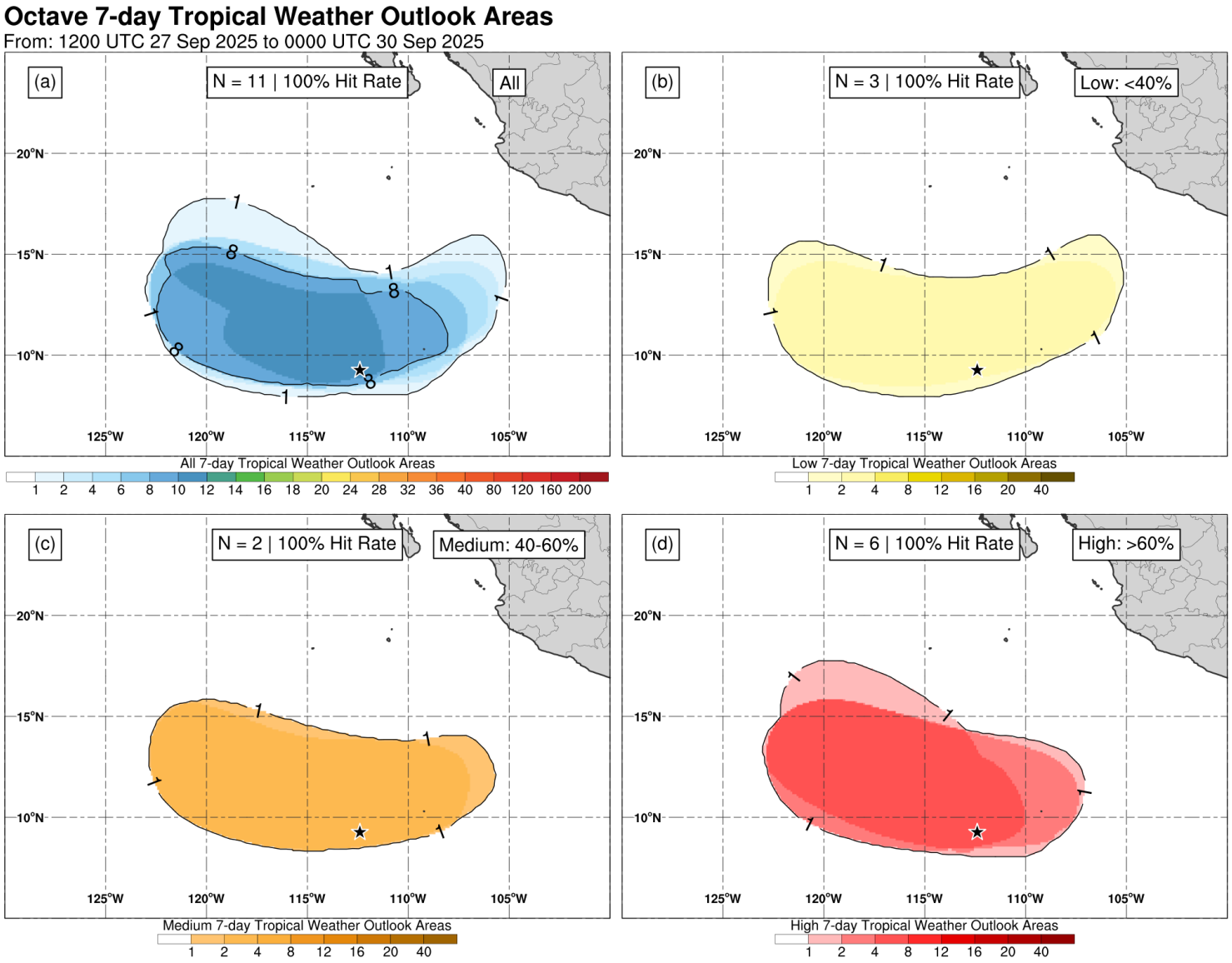


Figure 6. Composites of 7-day tropical cyclone genesis areas depicted in NHC's Tropical Weather Outlooks prior to the formation of Hurricane Octave, 30 September – 9 October 2025 for (a) all probabilistic genesis categories, (b) the low (<40%) category, (c) medium (40–60%) category, and (d) high (>60%) category. The location of genesis is indicated by the black star.